Chapter 1, Problem 5PP

Problem

Find the power delivered to the element in Example 1.5 at t = 5 ms if the current remains the same but the voltage is: (a) $v = 2i$ V, (b) $v = \left(10 + 5\int_0^t i\,dt\right)$ V.

Example 1.5:

Find the power delivered to an element at t = 3 ms if the current entering its positive terminal is

$$i = 5\cos 60\pi t \text{ A}$$

and the voltage is: (a) $v = 3i$, (b) $v = 3\,di/dt$.

Solution:

(a) The voltage is $v = 3i = 15\cos 60\pi t$; hence, the power is

$$p = vi = 75\cos^2 60\pi t \text{ W}$$

At t = 3 ms,

$$p = 75\cos^2(60\pi \times 3 \times 10^{-3}) = 75\cos^2 0.18\pi = 53.48 \text{ W}$$

(b) We find the voltage and the power as

$$v = 3\frac{di}{dt} = 3(-60\pi)5\sin 60\pi t = -900\pi \sin 60\pi t \text{ V}$$
$$p = vi = -4500\pi \sin 60\pi t \cos 60\pi t \text{ W}$$

At t = 3 ms,

$$\begin{aligned} p &= -4500\pi \sin 0.18\pi \cos 0.18\pi \text{ W} \\ &= -14137.167 \sin 32.4° \cos 32.4° = -6.396 \text{ kW} \end{aligned}$$

Step-by-step solution

Step 1 of 4

(a)

Consider the current in an element is $i = 5\cos 60\pi t$ A.

Consider the voltage across the element is,

$$\begin{aligned} v(t) &= 2i(t) \\ &= 2(5\cos 60\pi t) \\ &= 10\cos 60\pi t \end{aligned}$$

Step 2 of 4

Determine the power delivered to the element.

Consider the equation for the power.

$$\begin{aligned} p(t) &= v(t)i(t) \\ &= 10\cos 60\pi t \times 5\cos 60\pi t \\ &= 50\cos^2(60\pi t)\ \text{W} \end{aligned}$$

Substitute, $t = 5$ in equation.

$$\begin{aligned} p(t) &= 50\cos^2(60\pi t)\ \text{W} \\ &= 50\cos^2(60\pi(5))\ \text{W} \\ &= 50(0.5878)\ \text{W} \\ &= 17.27\ \text{W} \end{aligned}$$

Therefore, the power delivered to the element is $\boxed{17.27\ \text{W}}$.

Step 3 of 4

(b)

Consider the voltage across the element is,

$$v(t) = \left(10 + 5\int_0^t i\ dt\right)\ \text{V}$$

Replace $i = 5\cos 60\pi t\ \text{A}$ in equation.

$$\begin{aligned} v(t) &= 10 + 5\int_0^t 5\cos 60\pi t\ dt \\ &= 10 + 25\int_0^t \cos 60\pi t\ dt \\ &= 10 + 25\left[\frac{\sin 60\pi t}{60\pi}\right]_0^t \\ &= 10 + 25\left[\frac{\sin 60\pi t - \sin 0}{60\pi}\right] \end{aligned}$$

$$v(t) = 10 + 5\left[\frac{\sin 60\pi t}{12\pi}\right]\ \text{V}$$

Step 4 of 4

Determine the power delivered to the element.

Consider the equation for the power.

$$
\begin{aligned}
p(t) &= v(t)i(t) \\
&= \left[10 + 5\left[\frac{\sin 60\pi t}{12\pi}\right]\right] \times 5\cos 60\pi t \\
&= \left[50\cos 60\pi t + 25\left(\frac{2\cos 60\pi t \sin 60\pi t}{2\times 12\pi}\right)\right] \\
&= \left[50\cos 60\pi t + \frac{25}{24\pi}\sin 120\pi t\right] \text{ W}
\end{aligned}
$$

Substitute, $t = 5$ in equation.

$$
\begin{aligned}
p(t) &= \left[50\cos\left(60\pi \times 5\times 10^{-3}\right) + \frac{25}{24\pi}\sin\left(120\pi \times 5\times 10^{-3}\right)\right] \\
&= \left[50\cos(0.3\pi) + \frac{25}{24\pi}\sin(0.6\pi)\right] \\
&= \left[(50\times 0.5878) + \left(\frac{25}{24\pi}\times 0.9511\right)\right] \\
&= 29.7 \text{ W}
\end{aligned}
$$

Therefore, the power delivered to the element is $\boxed{29.7 \text{ W}}$.